

Vehicle Dynamics Simulation Model Overview

Vehicle-Dynamics-Sim-Can

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Abstract

This document provides a readable, end-to-end overview of the vehicle dynamics model implemented in the simulator. It is written for a general engineering audience and for readers who are new to vehicle dynamics simulation. The focus is on explaining model structure, the key physical assumptions, and the major tunable parameters. We also document the low-speed behavior fixes (kinematic blending and tire relaxation) and show representative simulation plots.

Contents

1 Purpose and Scope	2
2 System Architecture	2
3 State and Coordinate Definitions	2
4 Steering Model	2
5 Powertrain and Longitudinal Forces	3
6 Tire and Lateral Force Model	3
6.1 Tire Relaxation (First-Order Lag)	3
7 Vehicle Body Dynamics (3-DOF)	3
8 Low-Speed Stabilization	4
8.1 Kinematic Blend	4
8.2 Yaw Damping	4
9 Configuration and Tuning	4
10 Simulation Example	4
11 Tuning Guide (Quick Reference)	5
12 Glossary (Short)	5
13 Limitations and Future Work	6
14 Summary	6

1 Purpose and Scope

The simulator implements a lightweight vehicle dynamics model intended for closed-loop control testing and fast iteration. It is not a full-fidelity multi-body model; instead it uses a reduced-order 3-DOF rigid body model with simplified powertrain and tire behavior. This model is designed to be:

- Fast enough for real-time closed-loop simulation.
- Understandable and tunable through a YAML vehicle configuration.
- Stable for common test maneuvers and controller development.

2 System Architecture

The simulator is built as three primary subsystems that execute in a fixed order each timestep:

1. **SteerSubsystem:** Converts steer command into front-wheel angles using Ackermann geometry and rate limits.
2. **DriveSubsystem:** Converts torque/brake commands into longitudinal forces and lateral tire forces.
3. **VehicleSubsystem:** Integrates the vehicle body states (position, velocity, yaw) using forces from the drive subsystem.

Each subsystem updates a shared **PlantState**. The closed-loop controller sends commands over a virtual CAN bus and receives state feedback frames.

3 State and Coordinate Definitions

The vehicle model uses a body-fixed coordinate frame:

- x axis forward, y axis left, yaw positive counter-clockwise.
- Longitudinal velocity v (**v_mps**) and lateral velocity v_y (**vy_mps**) are expressed in the body frame.
- Yaw rate $\dot{\psi}$ (**yaw_rate_radps**) is around the vertical axis.

The global position (x, y) is obtained by rotating body velocities into the world frame each timestep.

4 Steering Model

The steering model uses:

- Ackermann geometry for inner/outer wheel angles.
- A steer angle rate limit (hydraulic steering response).

- Speed-based steering reduction.
- An additional low-speed steering clamp to prevent yaw oscillations.

The steering clamp is a simple linear ramp: the maximum steer angle is reduced at very low speeds and reaches full authority above a specified speed.

5 Powertrain and Longitudinal Forces

Longitudinal forces are computed from torque and brake commands:

- Motor torque is limited by maximum torque, power, drivetrain efficiency, and surface friction.
- Brake force is based on a maximum brake torque and a front/rear bias.
- Static rolling resistance and quadratic aerodynamic drag are applied.

These forces generate longitudinal acceleration, and the vehicle velocity is clamped to forward and reverse limits.

6 Tire and Lateral Force Model

Lateral forces use a linear tire model:

$$F_y = -C_y \alpha \quad (1)$$

where C_y is the cornering stiffness and α is the slip angle. To avoid instability near standstill, slip angles are scaled to zero as speed goes to zero, and a friction circle saturation clamps lateral forces.

6.1 Tire Relaxation (First-Order Lag)

To reduce high-frequency oscillations, lateral forces are passed through a first-order lag:

$$F_y(t + \Delta t) = F_y(t) + \alpha(F_{y,\text{target}} - F_y(t)) \quad (2)$$

where

$$\alpha = \frac{\Delta t}{\tau + \Delta t} \quad (3)$$

and τ is the tire relaxation time constant (configurable in YAML).

7 Vehicle Body Dynamics (3-DOF)

The vehicle body integrates longitudinal, lateral, and yaw dynamics. In body coordinates, the longitudinal acceleration is:

$$\dot{v}_x = \frac{F_x}{m} + v_y \dot{\psi} \quad (4)$$

The coupling term $v_y \dot{\psi}$ can dominate at low speed. This is why low-speed stabilization is essential.

8 Low-Speed Stabilization

Two key stabilizers are used:

8.1 Kinematic Blend

Below a configurable speed v_{blend} , yaw rate and lateral velocity are blended toward a kinematic bicycle model:

$$\dot{\psi}_{\text{kin}} = \frac{v}{L} \tan(\delta) \quad (5)$$

This prevents unrealistic oscillations when the vehicle is nearly stationary.

8.2 Yaw Damping

An additional low-speed yaw damping term is applied to reduce oscillations. This is a simple linear damping proportional to yaw rate.

9 Configuration and Tuning

Vehicle parameters are loaded from YAML files. Key tunables include:

- Mass, wheelbase, track width, CoG position, yaw inertia.
- Motor torque/power limits and brake torque.
- Rolling resistance and drag.
- Cornering stiffness values.
- **Low-speed kinematic blend threshold** `v_kinematic_blend_mps`.
- **Tire relaxation time constant** `tire_relax_tau_s`.

10 Simulation Example

Figure 1 shows a typical closed-loop run. The plots illustrate:

- Vehicle trajectory.
- Speed, longitudinal acceleration, and coupling term.
- Steering commands and yaw response.
- Motor torque and brake force.
- Wheel speeds.
- Lateral velocity and yaw rate.

Vehicle Dynamics Simulation — 3-DOF Plant

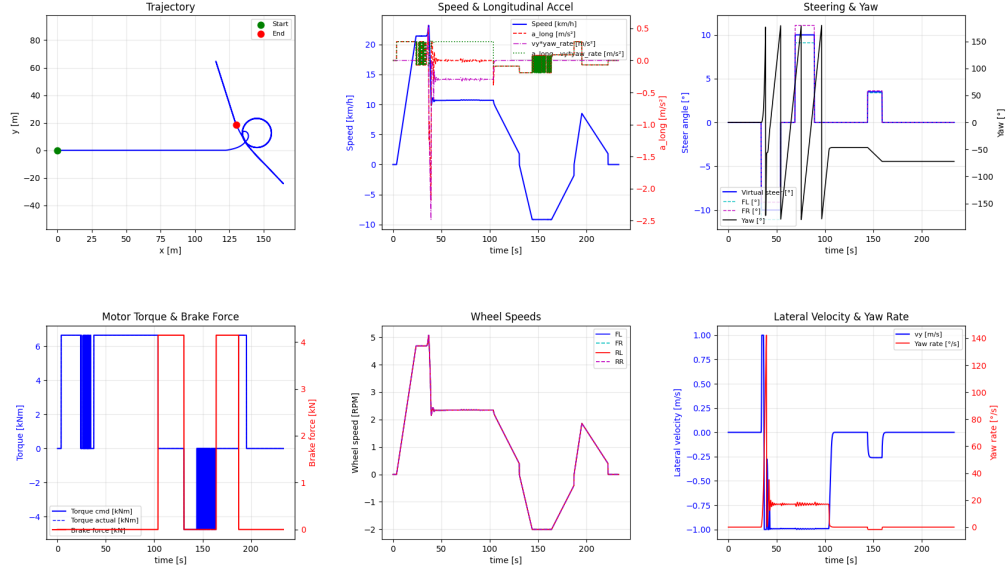


Figure 1: Closed-loop simulation results from `sim_out.csv`.

11 Tuning Guide (Quick Reference)

This section provides a short, practical tuning guide for common issues. Use it as a quick checklist before changing the controller.

- **Yaw oscillation at low speed:** increase `v_kinematic_blend_mps` or increase yaw damping.
- **Overly wide reverse turn:** increase `REV_STEER_DEG` or `Cy_rear_Npm`.
- **Harsh speed transitions:** increase tire relaxation time constant `tire_relax_tau_s`.
- **Excessive lateral drift:** lower `v_lat_max_mps` or reduce steer command.

Table 1: Key tuning parameters and suggested ranges

Parameter	Typical Range	Effect
<code>v_kinematic_blend_mps</code>	1.5–3.5	Low-speed stability and smooth turning.
<code>tire_relax_tau_s</code>	0.2–0.5	Reduces high-frequency oscillations.
<code>Cy_front_Npm</code>	2.0–3.0e6	Front lateral response / turn-in.
<code>Cy_rear_Npm</code>	2.0–3.2e6	Rear stability / understeer balance.
<code>yaw_inertia_kgm2</code>	6–9e6	Overall yaw responsiveness.
<code>REV_STEER_DEG</code>	15–25	Reverse turn tightness.

12 Glossary (Short)

- **Yaw:** Vehicle heading angle in the horizontal plane.

- **Yaw rate:** Rate of change of yaw (turning speed).
- **Slip angle:** Angle between wheel pointing direction and actual velocity.
- **Cornering stiffness:** Slope of lateral force vs. slip angle.
- **Kinematic bicycle:** A geometry-based turning model used at low speed.

13 Limitations and Future Work

This model is intentionally simplified. The main limitations are:

- Linear tire model (no combined-slip dynamics beyond friction circle clamp).
- No full wheel rotational dynamics or drivetrain inertia.
- No suspension or load transfer modeling.
- Simplified steering and brake models.

Potential future improvements include: richer tire models, more realistic steering compliance, and a flexible switch between kinematic and dynamic modes.

14 Summary

The simulator provides a practical balance between realism and simplicity. With the low-speed stabilizers, it produces stable closed-loop behavior suitable for controller development. The YAML configuration exposes the key knobs for tuning, while keeping the model approachable to non-specialists.